Xray for Jira Cloud

Step-by-step practical guide for manual testing, test organization, execution, and results

What this guide covers

- How to check that Xray Cloud is available in your Jira project
- Create Tests and write test steps
- Create and associate Preconditions
- Organize Tests with Test Sets and Test Plans
- Create Test Executions and record Test Run results
- Understand parameterization, evidence, and common troubleshooting

Important: Xray Cloud's menus and field names can vary by Jira layout, project type, permissions, and product updates. Treat the paths below as the usual workflow; if a named button is missing, use Jira search, the issue's More menu, or the Xray app navigation.

Prepared as a practical beginner reference

1. Understand the main Xray items

Item	What it means	When you use it
Test	A test case that describes what to check.	Write the scenario and its expected behavior.
Test Step	One action and the expected result within a manual Test.	Define the procedure a tester follows.
Precondition	A condition that must be true before a Test can run.	Record prerequisites such as a prepared account or test data.
Test Set	A collection of related Tests.	Group Tests by feature, team, or testing purpose.
Test Plan	A planning container for a testing scope or cycle.	Organize what should be tested for a release or milestone.
Test Execution	A record/container used to run selected Tests in a particular cycle.	Run tests and capture outcomes.
Test Run	The execution instance/result of a Test inside an execution.	Set status, record defects, and attach evidence.

2. Check that Xray Cloud is available

Before creating tests, confirm that Xray is installed for your Jira Cloud site and that your account has the required project permissions.

- Open the relevant Jira project.
- Look for Xray navigation, test-related project pages, or the Xray app in the Jira navigation.
- Try creating an issue and check whether issue types such as Test, Precondition, Test Set, Test Plan, and Test Execution are available.
- If those issue types or Xray screens are missing, ask your Jira administrator to confirm the Xray app installation, project configuration, issue-type scheme, and your permissions.

Permission note: You may be able to view tests but not create or edit them. Missing buttons do not always mean Xray is uninstalled.

3. Create a manual Test

Typical steps

- Open the Jira project where you want to store the test.
- Choose **Create** (or the project's equivalent create-issue action).
- Select the issue type **Test**.
- Enter a clear Summary, for example: *Verify that a user can log in with valid credentials*.
- If a Test Type field is available, choose **Manual**.
- Fill in required fields, then choose **Create**.
- Open the new Test issue and find its Test Details, Test Steps, or equivalent section/editor.

Write useful test steps

A manual step normally has an Action, optional Data, and an Expected Result. Some Xray Cloud layouts combine or label these fields differently.

Step	Action	Data	Expected result
1	Open the login page.	N/A	The login page is displayed.
2	Enter a valid username and password.	Valid test account	The credentials are accepted.
3	Select Log in.	N/A	The user reaches the home page.

Keep each step clear and observable. Avoid vague instructions such as “check everything.” State the action and the result that proves it passed.

4. Create a Precondition and associate it with a Test

Create the Precondition

- Choose **Create** and select issue type **Precondition** (or use the Xray Precondition creation option).
- Give it a descriptive Summary, such as *Valid user account exists*.
- Describe the setup requirement in the available details field, then create the issue.

Associate it with a Test

- Open the Test issue.
- Find the Precondition section, association control, or issue-linking area provided by your Xray Cloud layout.
- Search for the Precondition you created and add/associate it.
- Save the change if the interface requires it, then verify that the Precondition appears on the Test.

Example: Before testing a successful login, the tester needs an active account. Put that setup requirement in a Precondition rather than repeating long setup instructions in every login Test.

5. Create a Test Set and add Tests

- Create a Jira issue with type **Test Set**.
- Give it a useful name, such as *Login Regression Tests*.
- Open the Test Set and locate the Tests section or Add Tests action.
- Search for existing Tests, select the ones you want, and add them.
- Verify that the expected Tests are listed in the Test Set.

Use a Test Set to group related Tests. For example, one Test Set could contain login, logout, and password-reset Tests. A Test Set organizes tests; it does not itself record pass/fail results.

6. Create a Test Plan and add Tests

- Create a Jira issue with type **Test Plan**.
- Name it for the scope or cycle, for example *Release 2.4 Regression Plan*.
- Fill in any required project, version, label, or description fields.
- Open the Test Plan and find the Tests section or Add Tests action.
- Search for and add the Tests that belong in this plan.
- Review the plan to confirm it covers the intended testing scope.

A Test Plan answers: “What are we planning to test for this scope or release?” A Test Execution answers: “Which tests are we running in this particular run, and what were the results?” Depending on the workflow, executions can be linked to a plan.

7. Create a Test Execution and run Tests

Create the execution

- Create a Jira issue with type **Test Execution**.
- Give it a specific name, such as *Release 2.4 — Regression — Build 105*.
- Complete required fields and create the issue.
- Open the Test Execution and use its Tests/Add Tests control to select the Tests to run.

Add Tests from a Test Set

If the Test Execution screen provides an option to add Tests from a Test Set, use it and select the relevant set. If not, open the Test Set, select or identify its Tests, and add those individual Tests to the Test Execution. The exact workflow can vary by Xray Cloud interface version and permissions.

Execute and record results

- Open the Test Execution and select a Test Run or individual Test to start recording results.
- Follow the manual steps and compare actual behavior with each expected result.
- Set the overall Test Run status using the available status control, commonly PASS or FAIL. Other statuses may be available in your configuration.
- If a step fails, record what actually happened and attach a screenshot, log, or other evidence if useful.
- If a defect caused the failure, create or link the relevant Jira Bug using the available defect/linking control.
- Repeat for each Test in the execution and review the execution summary.

Important distinction: Adding a Test to an execution does not automatically mean it has passed. The Test Run result must be recorded after execution.

8. Parameterization and test data

Parameterization means running the same logical Test with different input values, such as several usernames or search terms. How you do this depends on your Xray Cloud test type, available features, and automation framework.

- For manual testing, you can create separate Test cases for distinct scenarios, or document a data table in the Test and execute each data row as required by your team's process.
- For automated testing, parameterization is usually implemented in the automation framework or test code. Xray then receives results through the supported integration/reporting format.
- Do not assume that a placeholder such as ${username} will automatically expand in every Xray Cloud Test. Confirm the syntax and feature support for your particular test type and integration.

Example test data:

Run	Username	Password	Expected result
1	valid_user	valid password	Login succeeds
2	locked_user	valid password	Login is denied with appropriate message
3	unknown_user	any password	Login is denied

9. Example end-to-end workflow

- **Write:** Create a Test called “Verify login with valid credentials” and add clear steps.
- **Prepare:** Create a Precondition called “Active test account exists” and associate it with the Test.
- **Organize:** Add the Test to a Test Set such as “Login Regression Tests.”
- **Plan:** Add the Test to a Test Plan for the release or testing cycle.
- **Run:** Create a Test Execution for a specific build and add the Test (directly or through the Test Set workflow available in your UI).
- **Record:** Execute the Test Run, set PASS or FAIL, add evidence, and link a defect if needed.
- **Review:** Check the execution summary and communicate remaining failures or unexecuted Tests.

10. Common problems and checks

Problem	What to check
Test or Precondition issue type is missing	Ask the Jira admin to check Xray installation, issue-type scheme, project configuration, and permissions.
Cannot edit Test Steps	Confirm the Test Type is Manual and that you have edit permission; look for Test Details or the Xray step editor.
Cannot associate a Precondition	Open the Test and look for the dedicated Precondition association section or supported link action; confirm permissions.
Tests do not appear in an execution	Check that the Tests exist in the same accessible project, then use the execution's Add Tests control and verify filters.
No PASS/FAIL control is visible	Open the Test Run from the Test Execution; result controls may not appear on the Test issue itself.
Automation results are not imported	Check the supported Xray Cloud import/reporting format, project permissions, Test issue keys, and integration configuration.

11. Quick checklist

- I can access Xray in my Jira Cloud project.
- I created a Manual Test with clear steps and expected results.
- I created and associated any required Precondition.
- I grouped related Tests in a Test Set.
- I added the intended Tests to a Test Plan.
- I created a Test Execution for the correct build/cycle.
- I ran each Test Run and recorded its actual result.
- I attached useful evidence and linked defects where appropriate.
- I reviewed the execution summary and unresolved failures.

12. Official documentation

Use the Xray Cloud documentation overview as the authoritative reference for current screens, permissions, test types, and automation integrations.

Because Xray Cloud is updated over time, verify exact menu labels and parameterization behavior in the documentation for your edition and setup.